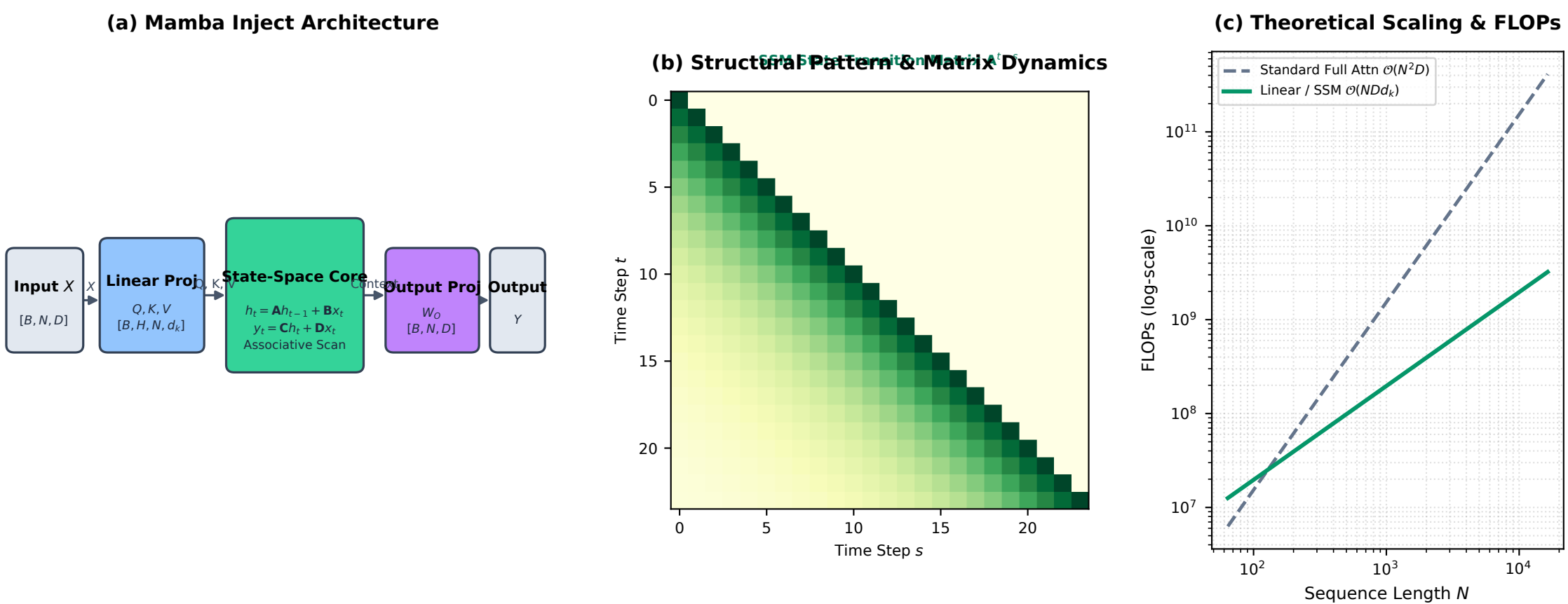
(a) Mamba Inject Architecture
Input $X$
$[B, N, D]$
$X$
Linear Proj
$Q, K, V$
$[B, H, N, d_k]$
$Q, K, V$
State-Space Core
$h_t = \mathbf{A}h_{t-1} + \mathbf{B}x_t$
$y_t = \mathbf{C}h_t + \mathbf{D}x_t$
Associative Scan
Context
Output Proj
$W_O$
$[B, N, D]$
Output
$Y$
(b) Structural Pattern & Matrix Dynamics
SSM State Transition Matrix $A^t$
Time Step $t$
Time Step $s$
0
5
10
15
20
(c) Theoretical Scaling & FLOPs
Standard Full Attn $\mathcal{O}(N^2D)$
Linear / SSM $\mathcal{O}(NDd_k)$
FLOPs (log-scale)
Sequence Length $N$
$10^7$
$10^8$
$10^9$
$10^{10}$
$10^{11}$
$10^2$
$10^3$
$10^4$